



PDZ-B series

Single Zener diodes

Rev. 5 — 21 December 2022

Product data sheet

1. General description

Low-power general purpose voltage regulator diodes in a very small SOD323 (SC-76) Surface-Mounted Device (SMD) plastic package.

2. Features and benefits

- Total power dissipation: $P_{\text{tot}} \leq 400 \text{ mW}$
- Small plastic package suitable for surface mounted design
- Wide variety of voltage ranges: nominal 2.4 V to 36 V (E24 range)
- Tolerance approximately $\pm 2 \%$
- PDZ5.1B - 10B: Very low dynamic impedances at low currents, very low leakage current, hard breakdown knee

3. Applications

- General voltage regulation

4. Quick reference data

Table 1. Quick reference data

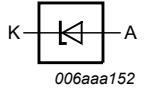
Symbol	Parameter	Conditions	Min	Typ	Max	Unit
V_F	forward voltage	$I_F = 10 \text{ mA}$ [1]	-	-	0.9	V
P_{tot}	total power dissipation	$T_{\text{amb}} \leq 25 \text{ °C}$ [2]	-	-	400	mW

[1] Pulse test: $t_p \leq 300 \text{ }\mu\text{s}$; $\delta \leq 0.02$.

[2] Device mounted on an FR4 Printed-Circuit Board (PCB), single-sided copper, tin-plated and standard footprint.

5. Pinning information

Table 2. Pinning

Pin	Symbol	Description	Simplified outline	Graphic symbol
1	K	cathode[1]		 006aaa152
2	A	anode		

[1] The marking bar indicates the cathode.

6. Ordering information

Table 3. Ordering information

Type number	Package		
	Name	Description	Version
PDZ2.4B to PDZ36B[1]	-	plastic surface-mounted package; 2 leads	SOD323

[1] The series consists of 29 types with nominal working voltages from 2.4 V to 36 V.

7. Marking

Table 4. Marking Codes

Type number	Marking Code	Type number	Marking Code	Type number	Marking Code
PDZ2.4B	Z0	PDZ6.2B	ZA	PDZ16B	ZL
PDZ2.7B	Z1	PDZ6.8B	ZB	PDZ18B	ZM
PDZ3.0B	Z2	PDZ7.5B	ZC	PDZ20B	ZN
PDZ3.3B	Z3	PDZ8.2B	ZD	PDZ22B	ZP
PDZ3.6B	Z4	PDZ9.1B	ZE	PDZ24B	ZQ
PDZ3.9B	Z5	PDZ10B	ZF	PDZ27B	ZR
PDZ4.3B	Z6	PDZ11B	ZG	PDZ30B	ZS
PDZ4.7B	Z7	PDZ12B	ZH	PDZ33B	ZT
PDZ5.1B	Z8	PDZ13B	ZJ	PDZ36B	ZU
PDZ5.6B	Z9	PDZ15B	ZK		

8. Limiting values

Table 5. Limiting values

In accordance with the Absolute Maximum Rating System (IEC 60134).

Symbol	Parameter	Conditions	Min	Max	Unit
I_F	continuous forward current		-	200	mA
I_{ZSM}	non-repetitive peak reverse current	$t_p = 100 \mu s$; square wave; $T_{amb} = 25^\circ C$ prior to surge	-	see characteristics table	
P_{tot}	total power dissipation	$T_{amb} = 25^\circ C$ [1]	-	400	mW
T_{stg}	storage temperature		-65	+150	$^\circ C$
T_j	junction temperature		-	+150	$^\circ C$

[1] Device mounted on an FR4 PCB, single-sided copper, tin-plated and standard footprint.

9. Thermal characteristics

Table 6. Thermal characteristics

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
$R_{th(j-sp)}$	thermal resistance from junction to solder point	in free air	-	-	130	K/W
$R_{th(j-a)}$	thermal resistance from junction to ambient	[1]	-	-	340	K/W

[1] Device mounted on an FR4 Printed-Circuit Board (PCB), single-sided copper, tin-plated and standard footprint.

10. Characteristics

Table 7. Characteristics

$T_j = 25^\circ C$ unless otherwise specified.

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
V_F	forward voltage	$I_F = 10 \text{ mA}$ [1]	-	-	0.9	V
V_F	forward voltage	$I_F = 100 \text{ mA}$ [1]	-	-	1.1	V

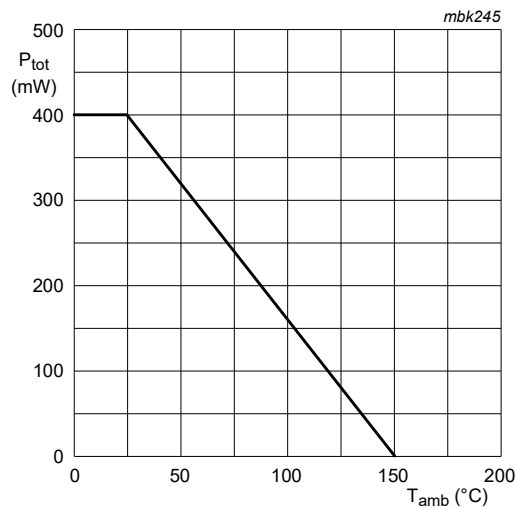
[1] Pulse test: $t_p \leq 300 \mu s$; $\delta \leq 0.02$.

Table 8. Characteristics per type; PDZ2.4B to PDZ36B

 $T_j = 25\text{ °C}$ unless otherwise specified.

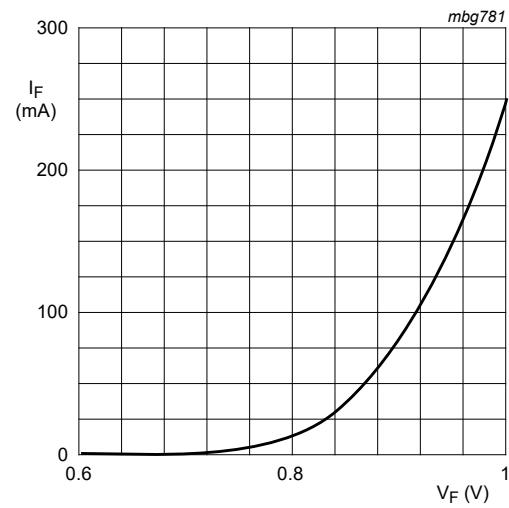
Type	Working voltage V_Z (V); $I_Z = 5\text{ mA}$		Maximum differential resistance r_{dif} (Ω)		Reverse current I_R (μA)		Temperature coefficient S_Z (mV/K); $I_Z = 5\text{ mA}$	Diode capacitance C_d (pF)[1]	Non-repetitive peak reverse current I_{ZSM} (A)[2]
	Min	Max	$I_Z = 0.5\text{ mA}$	$I_Z = 5\text{ mA}$	Max	V_R (V)	Typ	Max	Max
PDZ2.4B	2.43	2.63	1000	100	50	1.0	-1.6	450	8.0
PDZ2.7B	2.69	2.91	1000	100	20	1.0	-2.0	440	8.0
PDZ3.0B	2.85	3.07	1000	95	10	1.0	-2.1	425	8.0
PDZ3.3B	3.32	3.53	1000	95	5	1.0	-2.4	410	8.0
PDZ3.6B	3.60	3.85	500 @ 1 mA	90	5	1.0	-2.4	390	8.0
PDZ3.9B	3.89	4.16	500 @ 1 mA	90	3	1.0	-2.5	370	8.0
PDZ4.3B	4.17	4.48	600 @ 1 mA	90	3	1.0	-2.5	350	8.0
PDZ4.7B	4.55	4.75	600 @ 1 mA	90	2	1.0	-1.4	325	8.0
PDZ5.1B	4.96	5.20	250	60	2	1.5	0.3	300	5.5
PDZ5.6B	5.48	5.73	100	50	1	2.5	1.9	275	5.5
PDZ6.2B	6.06	6.33	80	50	0.5	3.0	2.7	250	5.5
PDZ6.8B	6.65	6.93	60	40	0.5	3.5	3.4	215	5.5
PDZ7.5B	7.28	7.60	60	10	0.5	4.0	4.0	170	3.5
PDZ8.2B	8.02	8.36	60	10	0.5	5.0	4.6	150	3.5
PDZ9.1B	8.85	9.23	60	10	0.5	6.0	5.5	120	3.5
PDZ10B	9.77	10.21	60	10	0.1	7.0	6.4	110	3.5
PDZ11B	10.78	11.22	60	10	0.1	8.0	7.4	108	3.0
PDZ12B	11.74	12.24	80	10	0.1	9.0	8.4	105	3.0
PDZ13B	12.91	13.49	80	10	0.1	10.0	9.4	103	2.5
PDZ15B	14.34	14.98	80	15	0.05	11.0	11.4	99	2.0
PDZ16B	15.85	16.51	80	20	0.05	12.0	12.4	97	1.5
PDZ18B	17.56	18.35	80	20	0.05	13.0	14.4	93	1.5
PDZ20B	19.52	20.39	100	20	0.05	15.0	16.4	88	1.5
PDZ22B	21.54	22.47	100	25	0.05	17.0	18.4	84	1.3
PDZ24B	23.72	24.78	120	30	0.05	19.0	20.4	80	1.3
PDZ27B	26.19	27.53	150	40	0.05	21.0	23.4	73	1.0
PDZ30B	29.19	30.69	200	40	0.05	23.0	26.6	66	1.0
PDZ33B	32.15	33.79	250	40	0.05	25.0	29.7	60	0.9
PDZ36B	35.07	36.87	300	60	0.05	27.0	33.0	59	0.8

[1] $f = 1\text{ MHz}$; $V_R = 0\text{ V}$.[2] $t_p = 100\text{ }\mu\text{s}$; $T_{amb} = 25\text{ °C}$.



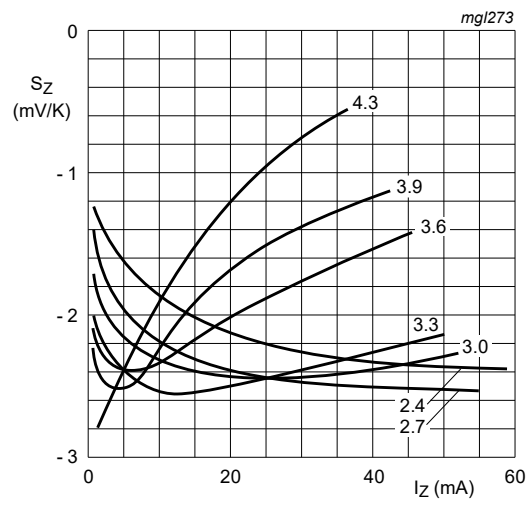
FR4 PCB, standard footprint

Fig. 1. Power derating curve



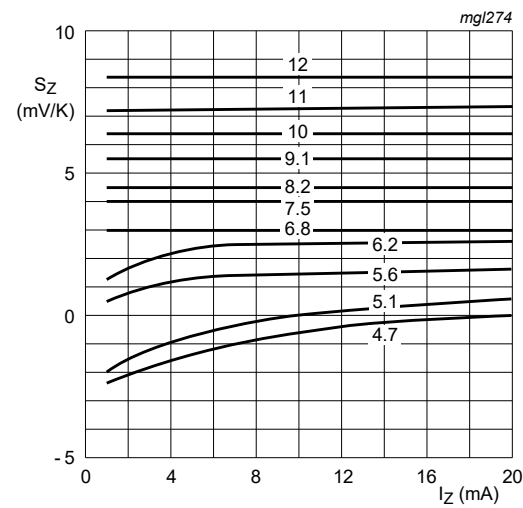
$T_j = 25\text{ °C}$

Fig. 2. Forward current as a function of forward voltage; typical values



PDZ2.4B to PDZ4.3B
 $T_j = 25\text{ °C}$ to 150 °C

Fig. 3. Temperature coefficient as a function of working current; typical values



PDZ4.7B to PDZ12B
 $T_j = 25\text{ °C}$ to 150 °C

Fig. 4. Temperature coefficient as a function of working current; typical values

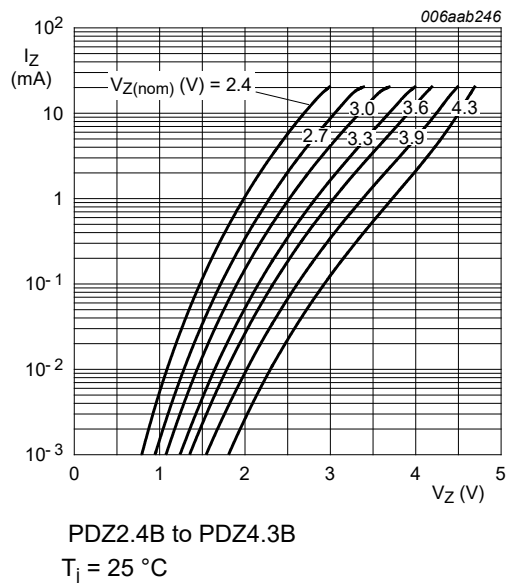


Fig. 5. Working current as a function of working voltage; typical values

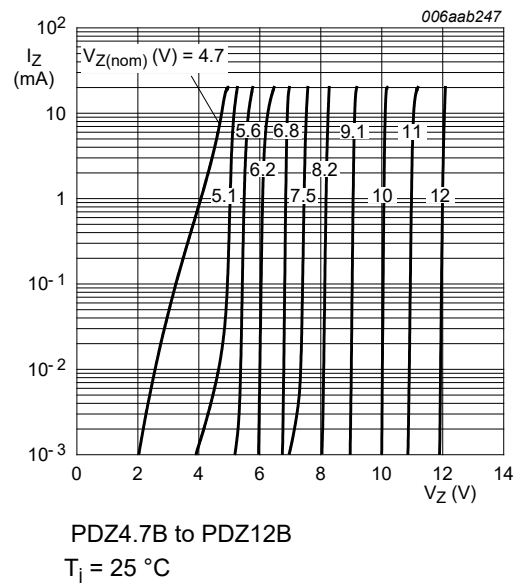


Fig. 6. Working current as a function of working voltage; typical values

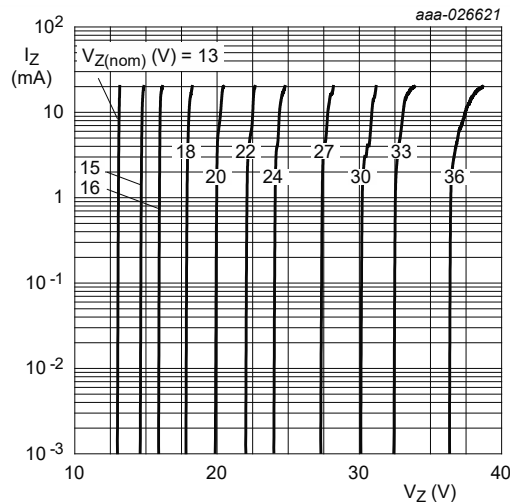


Fig. 7. Working current as a function of working voltage; typical values

11. Package outline

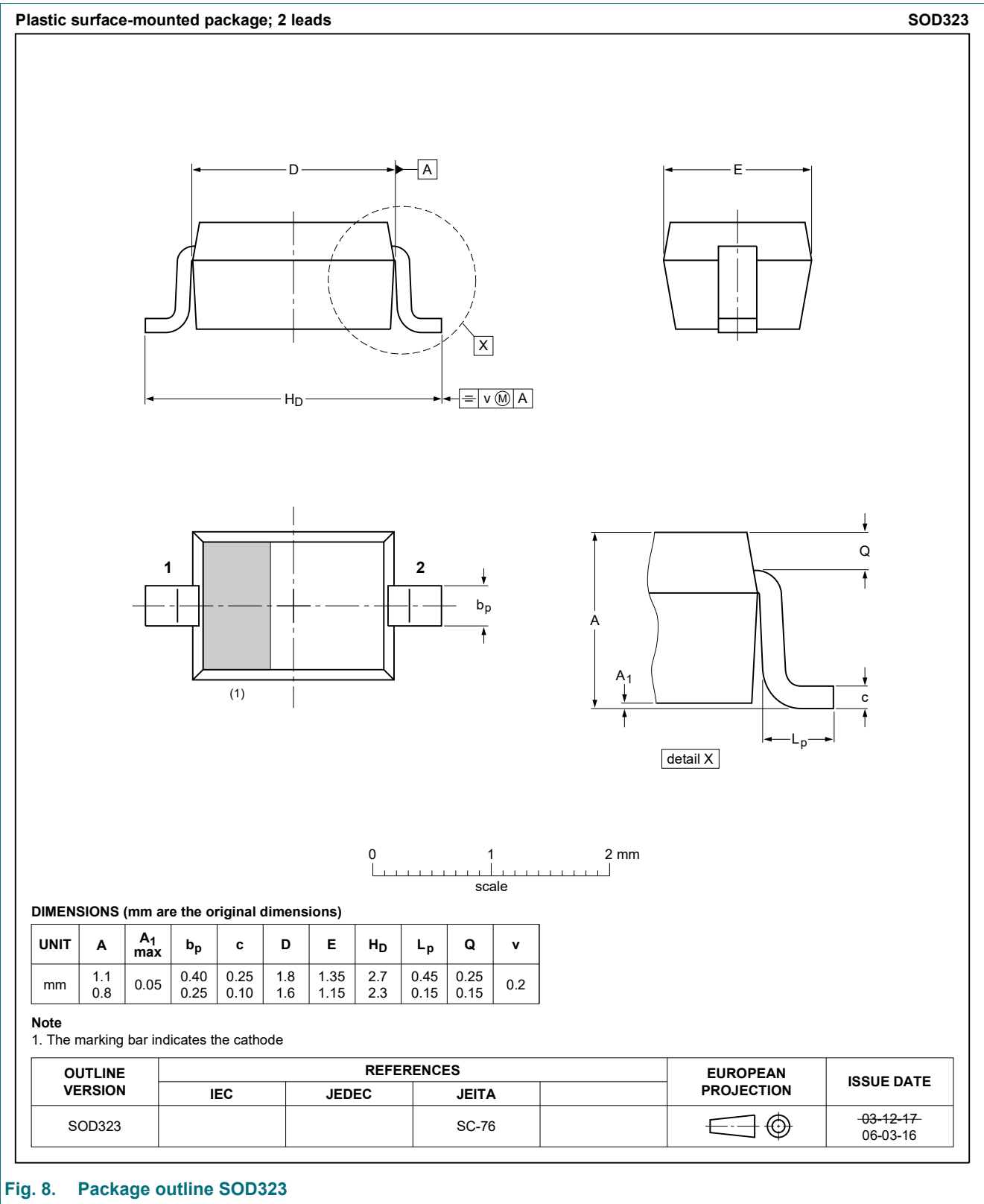


Fig. 8. Package outline SOD323

12. Soldering

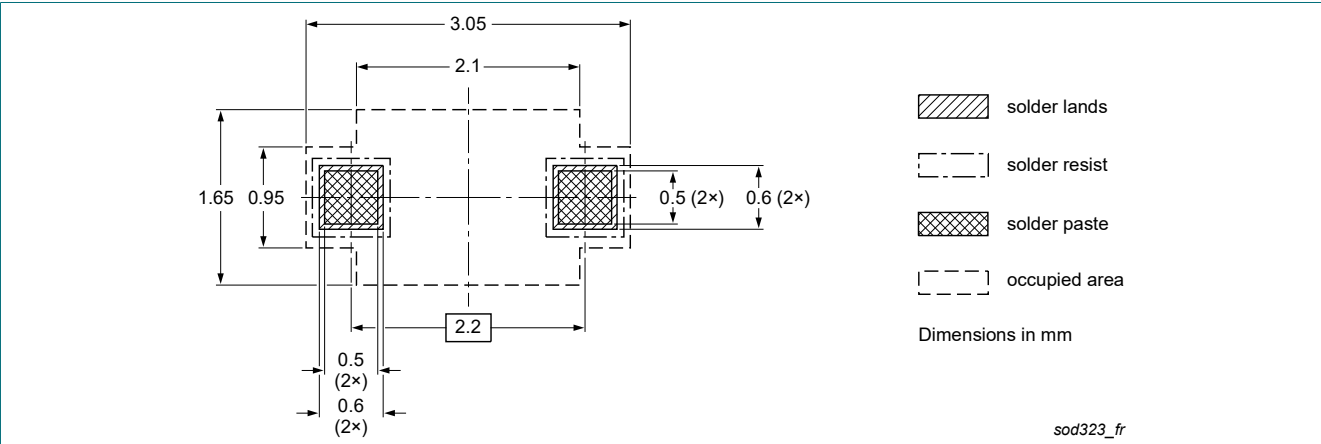


Fig. 9. Reflow soldering footprint SOD323

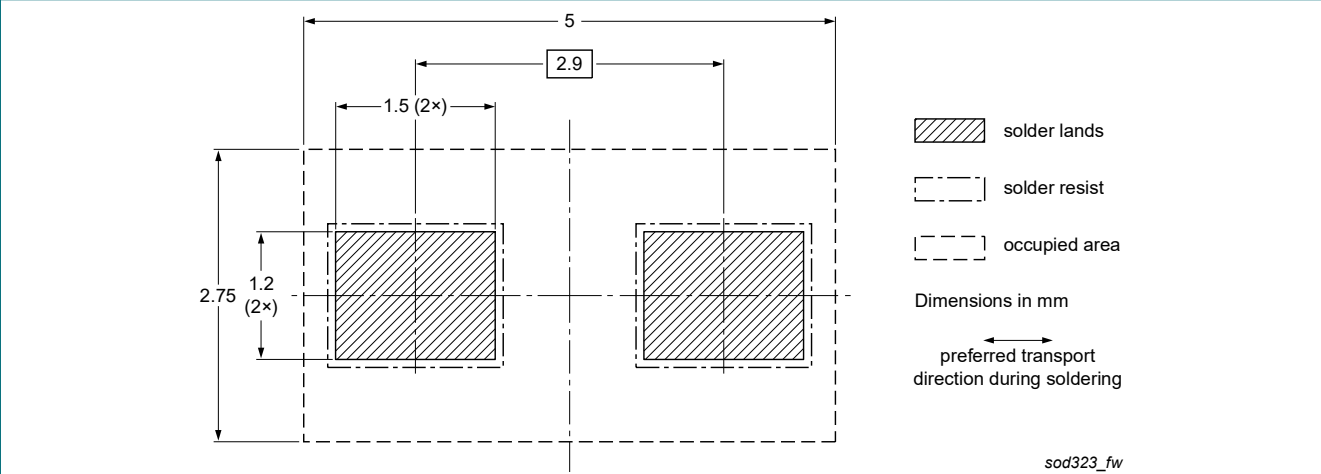


Fig. 10. Wave soldering footprint SOD323

13. Revision history

Table 9. Revision history

Document ID	Release date	Data sheet status	Change notice	Supersedes
PDZ-B_SER v.5	20221221	Product data sheet	-	PDZ-B_SER v.4
Modifications:	• Figure 1: Notes and title corrected			
PDZ-B_SER v.4	20220701	Product data sheet	-	PDZ-B_SER v.3
PDZ-B_SER v.3	20190305	Product data sheet	-	PDZ-B_SER v.2
PDZ-B_SER v.2	20040322	Product data sheet	-	PDZ-B_SER v.1
PDZ-B_SER v.1	20020218	Product data sheet	-	-

14. Legal information

Data sheet status

Document status [1][2]	Product status [3]	Definition
Objective [short] data sheet	Development	This document contains data from the objective specification for product development.
Preliminary [short] data sheet	Qualification	This document contains data from the preliminary specification.
Product [short] data sheet	Production	This document contains the product specification.

- [1] Please consult the most recently issued document before initiating or completing a design.
- [2] The term 'short data sheet' is explained in section "Definitions".
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